بسم الله الرحمن الرحيم.

Section (5) Chemistry.

- Aldehyde [R - C(=O) - H]

1). Aliphatic
- HCHO, Formaldehyde.
- CH_3CHO, Acetaldehyde.

2). Aromatic

- Benzaldehyde.

* Chemical Properties:

1). Shiff's reagent:
- 1ml of unknown liquid.
- 1ml of Shiff's reagent.

observation ~~Result~~ → Pink Red Color.

تفاعلات بجعلها عشان أعرف ده الوحيد أو لا

2). Ammonical mirror image test:
- 1ml of unknown liquid.
- 5ml of ammonical $AgNO_3$.
- water bath for 15ml

→ mirror image
→ بقعة زيتية
Black ppt.

MALEK

3). Fehling test:

1) 1ml of Fehling A.
2) 2ml ~ ____ B.
3) Shaking ثم الهز.
4) 1ml of Na_2CO_3.
5) 1ml of unknown liquid
6) water bath of (5-10)m

→ Red P.P.t.

متفاعلات بعملها عشان أعرف ده فورمالدهيد أستيالدهيد:

- miscibility → miscible → . formaldehyde - Acetaldehyde.
  - → immiscible → . Benzaldehyde.

Analine → immisible, Brown. نخلي بالنا.

1) Aldol Condensation test:

1ml of aldehyde
(3-5) ml of Conc NaOH
water bath (5-20)m

→ Yellow → orange P.P.t.

. acetaldehyde. ← +Ve لو.
. formaldehyde. ← -Ve لو.

2). Iodo form test:

1 ml of aldehyde
3 ml of I_2
Drops of dil NaoH

→ Brown → Yellow → Yellow P.P.t on cold.

- acsetaldehyde. ← +Ve ل.
- formaldehyde. ← -Ve ل.

3). Cannazore test:

[3-5] ml of conc NaoH
1 ml of aldehyde
Bath water for [5-10] m ater cooling
add 3 ml of conc HCl

→ white P.P.t Benzealdehyde